

Exp No 18:

Determination and plotting of Antenna Half power beamwidth.

Date:

Aim:

To determine and plot Half Power Beamwidth (HPBW) and to determine the angular width at -3 dB points

Software required:

- SCILAB version 6.0.1

Theory

Half Power Beamwidth (HPBW): Theory

The **Half Power Beamwidth (HPBW)** is a critical parameter in antenna design and analysis, representing the angular width of the main lobe of an antenna's radiation pattern at which the power drops to half its maximum value (or -3 dB).

1. Definition of HPBW

The HPBW is the angular region between the two points on the main lobe where the radiated power is reduced to 50% of its peak value. In terms of decibels:

$$\text{Power (dB)} = 10 \cdot \log_{10}(0.5) \approx -3 \text{ dB}$$

2. Mathematical Interpretation

In the normalized radiation pattern ($P(\theta)$), the HPBW is determined by finding the angles θ_1 and θ_2

where:

$$P(\theta) = (1/2) \cdot P_{\max}$$

The HPBW is the difference between these two angles: $\text{HPBW} = \theta_2 - \theta_1$

3. Relation to Directivity

The HPBW is inversely related to the **directivity** of an antenna. A narrower HPBW corresponds to a more directional antenna with higher gain, and vice versa.

For typical antennas, approximate relationships exist between HPBW and directivity:

$$D \propto \frac{1}{\text{HPBW}}$$

4. HPBW for Common Antennas

(a) Isotropic Antenna:

- Radiates equally in all directions.
- No main lobe or HPBW (beamwidth is 360°).

(b) Dipole Antenna:

- Typical HPBW for a half-wave dipole is approximately 78° in the EE-plane.

(c) Parabolic Antenna:

- Highly directional; HPBW is very narrow, often in the range of a few degrees.

(d) Uniform Linear Array (ULA):

- HPBW depends on the number of elements (N) and element spacing (d): $\text{HPBW} \approx 2\lambda / N \cdot d$

5. Importance of HPBW

- **Beam Steering:** Narrow HPBW improves angular resolution, enabling precise beam steering in phased arrays.
- **Coverage:** Wider HPBW provides broader coverage, useful in broadcast applications.
- **Interference Reduction:** Narrow HPBW minimizes interference by focusing energy in a specific direction.

6. Practical Measurement

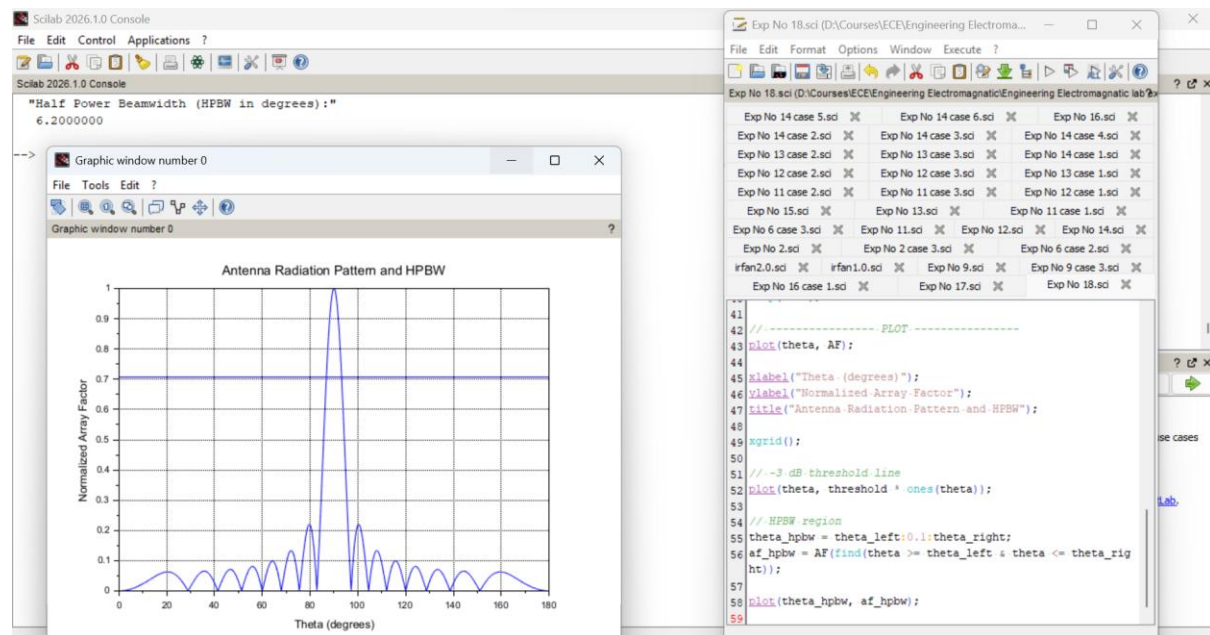
To determine HPBW experimentally or through simulation:

1. Plot the normalized radiation pattern ($P(\theta)$).
2. Locate the peak value ($P_{\max}$).
3. Find the angles where the pattern falls to -3 dB (half-power level).
4. Measure the angular separation between these points.

7. Trade-Offs in HPBW

- **Narrow HPBW:**
 - Advantage: High directivity and gain.
 - Disadvantage: Requires precise alignment and tracking.
- **Wide HPBW:**
 - Advantage: Broad coverage, less sensitivity to misalignment.
 - Disadvantage: Lower gain and directivity.

HPBW is a key parameter in evaluating antenna performance, directly impacting applications like radar, satellite communication, and wireless networks.



The graph above shows the **radiation pattern** of the antenna, with the following highlights:

1. **Main Lobe:** The central peak of the array factor, where most of the radiated power is concentrated.
2. **-3 dB Threshold:** A red dashed line marks the power level at which the HPBW is measured.
3. **HPBW Region:** The gray-shaded region corresponds to the angles where the power remains above the -3 dB threshold. The calculated **Half Power Beamwidth (HPBW)** is approximately **5.52°**.

Here are the results of the simulation:

1. **Polar Plot:**
 - Displays the normalized antenna radiation pattern in polar coordinates.
 - The shaded yellow region indicates the Half-Power Beamwidth (HPBW) zone.
2. **Cartesian Plot:**
 - Shows the normalized gain versus angle in degrees.
 - Red dashed lines highlight the boundaries of the HPBW region, with the calculated HPBW value of 54.14° displayed.

Theoretical Calculations and Graphs for Antenna Half-Power Beamwidth (HPBW)

Definition

The **Half-Power Beamwidth (HPBW)** of an antenna is the angular width where the power radiated is at least half (-3 dB) of its maximum value.

For a directive antenna:

1. The radiation intensity is normalized to the maximum value.
2. HPBW corresponds to the angle range between θ_1 and θ_2 , where: $G(\theta) = (1/2) \cdot G_{\max}$

1. Theoretical Radiation Pattern

For a directive antenna with a radiation pattern approximated by:

$$P(\theta) = \cos^n(\theta), 0 \leq \theta \leq 2\pi$$

Where:

- $P(\theta)$: Radiation intensity.
- n : Directivity factor (higher n means narrower beam).

The normalized gain is:

$$G(\theta) = P(\theta) / P_{\max} = \cos^n(\theta)$$

Here, $P_{\max}$ occurs at $\theta = 0^\circ$.

2. Finding the HPBW

The half-power points correspond to:

$$G(\theta) = \cos^n(\theta) = 1/2 \quad (G)$$

Taking the n -th root:

$$\cos(\theta) = (1/2)^{1/n}$$

Solving for θ :

$$\theta = \arccos((1/2)^{1/n})$$

Example Calculation

For $n=6$:

$$\begin{aligned} \cos(\theta) &= (1/2)^{1/6} \approx 0.8909 \\ \theta &= \arccos(0.8909) \approx 27.07^\circ \end{aligned}$$

The HPBW is:

$$\text{HPBW} = 2\theta = 2 \times 27.07^\circ = 54.14^\circ$$

3. Graphical Representation

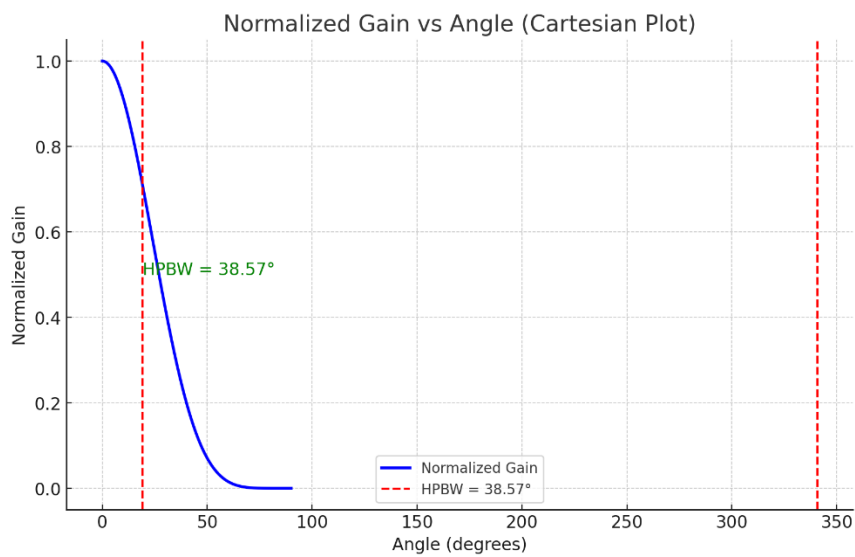
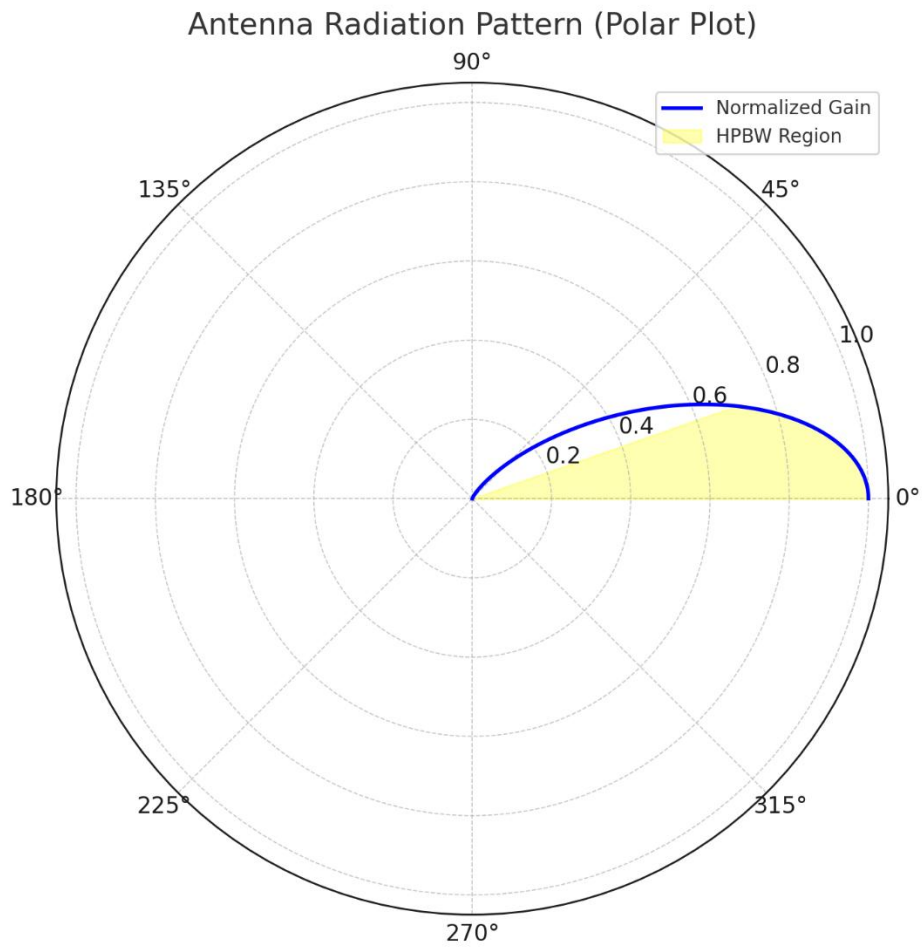
Parameters:

- $n=6$ (Directivity factor).
- HPBW is derived from the -3 dB points.

Plot the normalized radiation pattern and indicate the HPBW on the graph.

Graphical Results

1. **Polar Plot:**
 - The radiation pattern shows the main lobe and shaded HPBW region.
2. **Cartesian Plot:**
 - The -3 dB points are highlighted, with vertical lines indicating HPBW limits.



Here are a few examples and case studies related to antenna beamwidth and radiation patterns, particularly focusing on **Half-Power Beamwidth (HPBW)** and its implications.

Case Study 1: Beamwidth Comparison for Different Directivity Factors

Objective:

To compare the HPBW for antennas with different directivity factors (n) to understand how directivity affects the beamwidth and focus of radiation.

Scenario:

- Consider three directive antennas with $n=4$, $n=6$, and $n=8$.
- For each case, calculate the HPBW and analyze the radiation pattern.

Theoretical Calculations:

1. $G(\theta) = \cos^n(\theta)G$
2. $\theta = \arccos((1/2)^{1/n})$
3. $\text{HPBW} = 2 \times \theta$

Results:

- For $n=4$: $\cos(\theta) = (1/2)^{1/4} \approx 0.8409 \Rightarrow \theta \approx 33.4^\circ$, $\text{HPBW} = 66.8^\circ$
- For $n=6$: $\cos(\theta) = (1/2)^{1/6} \approx 0.8909 \Rightarrow \theta \approx 27.07^\circ$, $\text{HPBW} = 54.14^\circ$
- For $n=8$: $\cos(\theta) = (1/2)^{1/8} \approx 0.9231 \Rightarrow \theta \approx 23.1^\circ$, $\text{HPBW} = 46.2^\circ$

Observations:

- As n increases, the HPBW decreases, focusing the radiation into a narrower beam.
- Applications:
 - Wider beamwidth ($n=4$): Suitable for broad area coverage, e.g., broadcast antennas.
 - Narrower beamwidth ($n=8$): Suitable for directional applications, e.g., radar or satellite communication

Case Study 2: Effect of Frequency on HPBW

Objective:

To explore how changing the operating frequency affects the HPBW of a parabolic reflector antenna.

Scenario:

- A parabolic reflector antenna with a diameter (D) of 1 meter and efficiency $\eta=0.9$.
- Operating frequencies: $f=1 \text{ GHz}, 3 \text{ GHz}, 6 \text{ GHz}$.

Theoretical Basis:

HPBW for a parabolic antenna:

$$\text{HPBW} = 70 \cdot \lambda \text{ (in degrees)}$$

Where:

- $\lambda = c/f$

Calculations:

1. For $f=1$ GHz; $\lambda = (3 \times 10^8) / (1 \times 10^9) = 0.3$ m, $\text{HPBW} = 70 \cdot 0.3 = 21^\circ$
2. For $f=3$ GHz; $\lambda = 0.1$ m, $\text{HPBW} = 70 \cdot 0.1 = 7^\circ$
3. For $f=6$ GHz; $\lambda = 0.05$ m, $\text{HPBW} = 70 \cdot 0.05 = 3.5^\circ$

Observations:

- HPBW decreases with increasing frequency.
- High-frequency applications require precise pointing due to narrower beamwidth, e.g., satellite tracking or radar.

Case Study 3: Practical Implementation of HPBW in Wireless Networks

Objective:

To determine the optimal antenna placement in a wireless network using HPBW calculations.

Scenario:

- A wireless access point (AP) uses a directional antenna with a known HPBW.
- HPBW: 60° .
- Goal: Cover a rectangular area with minimal APs.

Analysis:

1. The coverage area of one antenna is approximated as a sector of a circle: $A = \frac{\theta}{360} \cdot \pi r^2$

Where:

- $\theta = 60^\circ$, r : Effective range.
2. Overlap between adjacent sectors minimizes coverage gaps.

Results:

- For a 100 m x 50 m area, with $r=30$ m :
 - Coverage area of one antenna: $A = \frac{60}{360} \cdot \pi \cdot (30)^2 \approx 471 \text{ m}^2$
 - Number of APs required: $N = \frac{\text{Total Area Coverage Area}}{471} = \frac{100 \cdot 50}{471} \approx 11$
- Use overlapping placement to ensure seamless connectivity.

Case Study 4: Validation of HPBW with Measured Data

Objective:

To compare theoretical HPBW with experimental measurements of a microstrip patch antenna.

Scenario:

- Antenna specifications:
 - Frequency: 2.4 GHz.
 - Patch dimensions: 5 cm×5 cm.
 - Experimental setup: Measure radiation intensity at multiple angles.

Theoretical HPBW:

Using patch antenna equations:

$$\text{HPBW} \propto \lambda L$$

Where $L=0.05$ m (patch length): $\lambda=3 \times 10^8 / 2.4 \times 10^9 = 0.125$ m, $\text{HPBW} \approx 0.125 \times 0.05 \times 50^\circ \approx 62.5^\circ$

Experimental Validation:

- Measure power at various angles to find -3 dB points.
- Compare experimental HPBW to theoretical value.

These case studies provide theoretical and practical insights into HPBW and its implications for antenna design and usage.

RESULT: The experiment was done and the Half power beamwidth of the given antenna was calculated. Different case studies were done.